

|                                     | FaultExplainer<br>GPT-4o | FaultExplainer<br>o1-preview | PCA<br>detectable | IDD (this work)<br>no candidates |
|-------------------------------------|--------------------------|------------------------------|-------------------|----------------------------------|
| IDV1 A/C feed-ratio step            | wrong                    | correct                      | yes               | Top-1 (0.88)                     |
| IDV3 D-feed temperature step (hard) | unscored                 | unscored                     | no                | capped CS (0.60)                 |
| IDV4 reactor cooling-water step     | unscored                 | unscored                     | no                | Top-1 (0.75)                     |
| IDV7 C-header pressure loss         | correct                  | correct                      | yes               | Top-1 (0.82)                     |
| IDV11 cooling-water random          | correct                  | correct                      | yes               | Top-1 (0.80)                     |
| IDV14 cooling-valve stiction        | correct                  | correct                      | yes               | Top-1 (0.82)                     |